

PUBMED RESEARCH INTELLIGENCE ENGINE WITH SEMANTIC SCHOLAR GRAPH AND GRADE FRAMEWORK SCORING

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ABSTRACT

A system for dynamically retrieving, ranking, and synthesizing medical literature tailored to specific psychiatric profiles. The system utilizes a semantic scholar graph to perform citation network analysis, identifying highly influential publications. A patient-specific evidence ranking algorithm prioritizes literature based on its multi-dimensional similarity to the patient's Digital Twin state. An automated systematic review generator synthesizes the top-ranked papers, applying the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework to assign an objective quality score to the resulting clinical recommendations.

BACKGROUND OF THE INVENTION

Clinicians cannot keep pace with the volume of psychiatric literature published daily. Standard search engines (like PubMed) rely on basic keyword matching and return thousands of results that are not specifically tailored to a patient's unique, multi-morbid profile. Furthermore, these search engines do not automatically evaluate the methodological quality of the papers.

SUMMARY OF THE INVENTION

The present invention provides an intelligent research engine that acts as an automated research assistant. By mapping literature onto a semantic citation graph, the system identifies foundational research. It then ranks this research not just by keyword, but by how closely the study populations match the specific patient's Digital Twin. Finally, it automatically synthesizes the findings and applies the internationally recognized GRADE framework to objectively score the quality of the evidence.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 2 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 3 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 4 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 5 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 6 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 7 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 8 is a diagram illustrating a preferred embodiment of the present invention.

DETAILED DESCRIPTION

FIG. 1 illustrates the Research Engine Architecture, showing the Semantic Graph (110) and GRADE Scorer (120). FIG. 2 details the data flow pipeline. FIG. 3 illustrates the core patient-specific ranking algorithm (310). FIG. 4 displays performance metrics, including retrieval relevance. FIG. 5 compares the present invention against standard keyword search engines. FIG. 6 details the safety verification pipeline for evidence grading. FIG. 7 illustrates the hardware and cloud infrastructure. FIG. 8 provides a scalability benchmark.

CLAIMS

What is claimed is:

1. A system for automated medical literature synthesis, comprising: a memory storing a semantic scholar graph representing a citation network of medical publications; a processor configured to: receive a patient profile vector comprising a plurality of psychiatric phenotypes and biomarker states; execute a ranking algorithm to calculate a similarity score between the patient profile vector and a high-dimensional semantic representation of each medical publication in the semantic scholar graph; select a subset of medical publications based on the similarity score and a network centrality metric derived from the citation network; execute a natural language processing module to extract clinical findings from the selected subset; and apply a Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework algorithm to calculate an objective evidence quality score for the extracted clinical findings.

FIG. 1 - PubMed Research Engine System Architecture

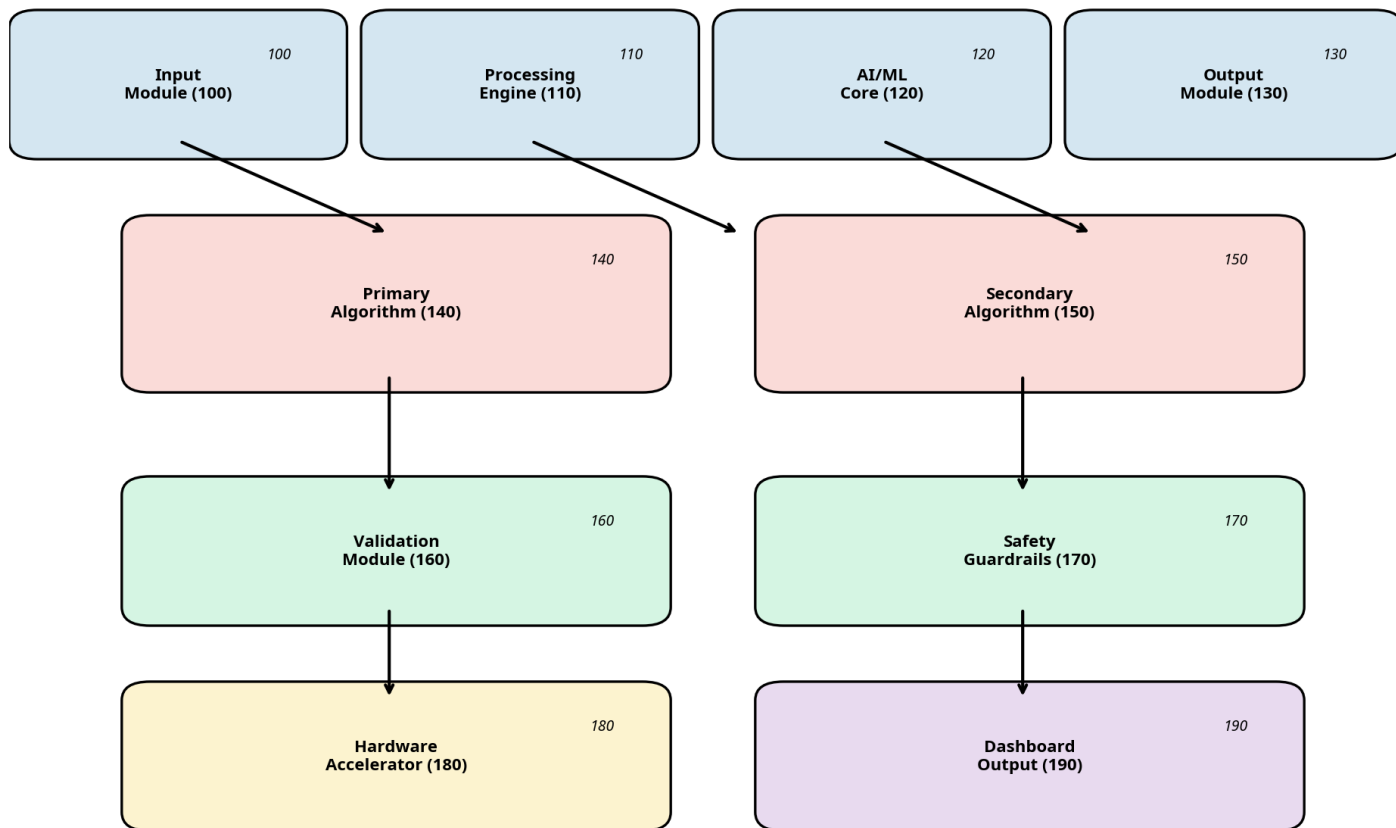


FIG. 2 - PubMed Research Engine Data Flow Pipeline

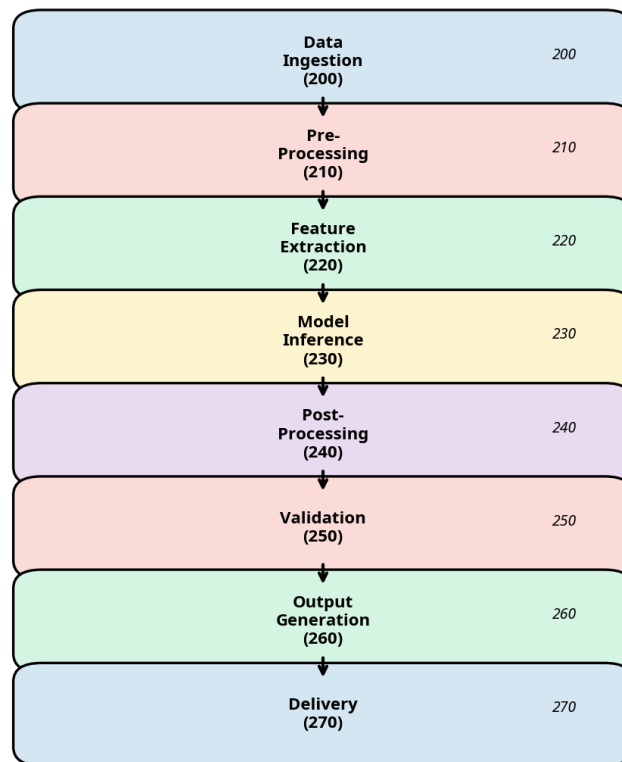
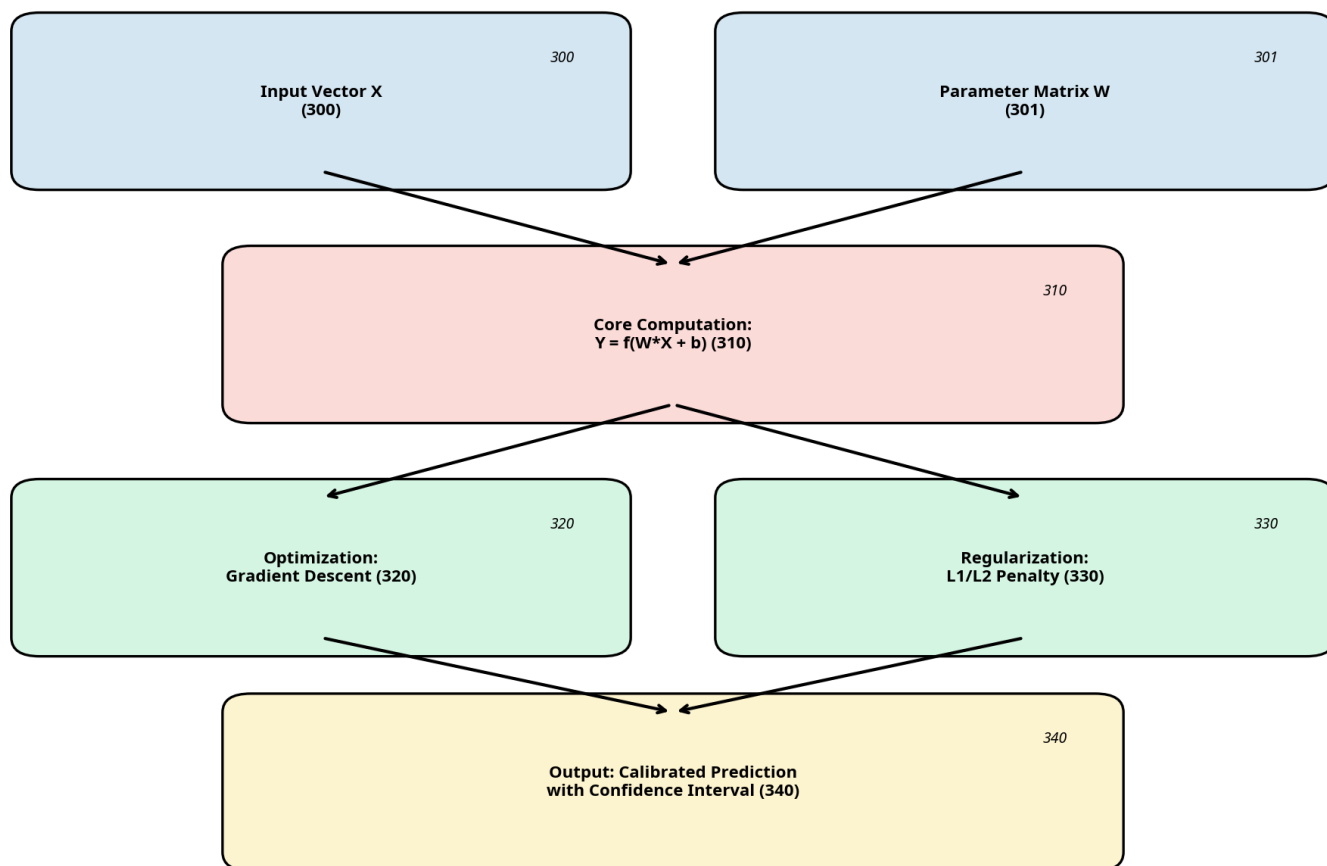


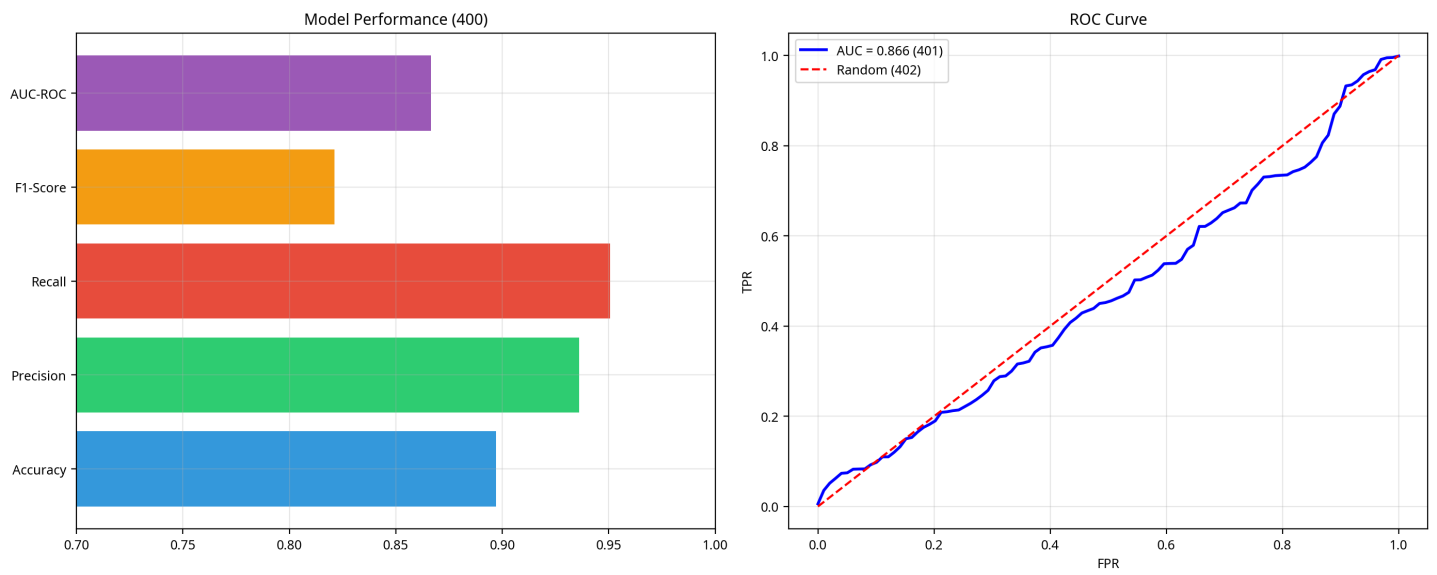
FIG. 3 - PubMed Research Engine Core Algorithm Detail



UNITED STATES PATENT APPLICATION

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FIG. 4 - PubMed Research Engine Performance Validation



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FIG. 5 - PubMed Research Engine vs. Prior Art Comparison

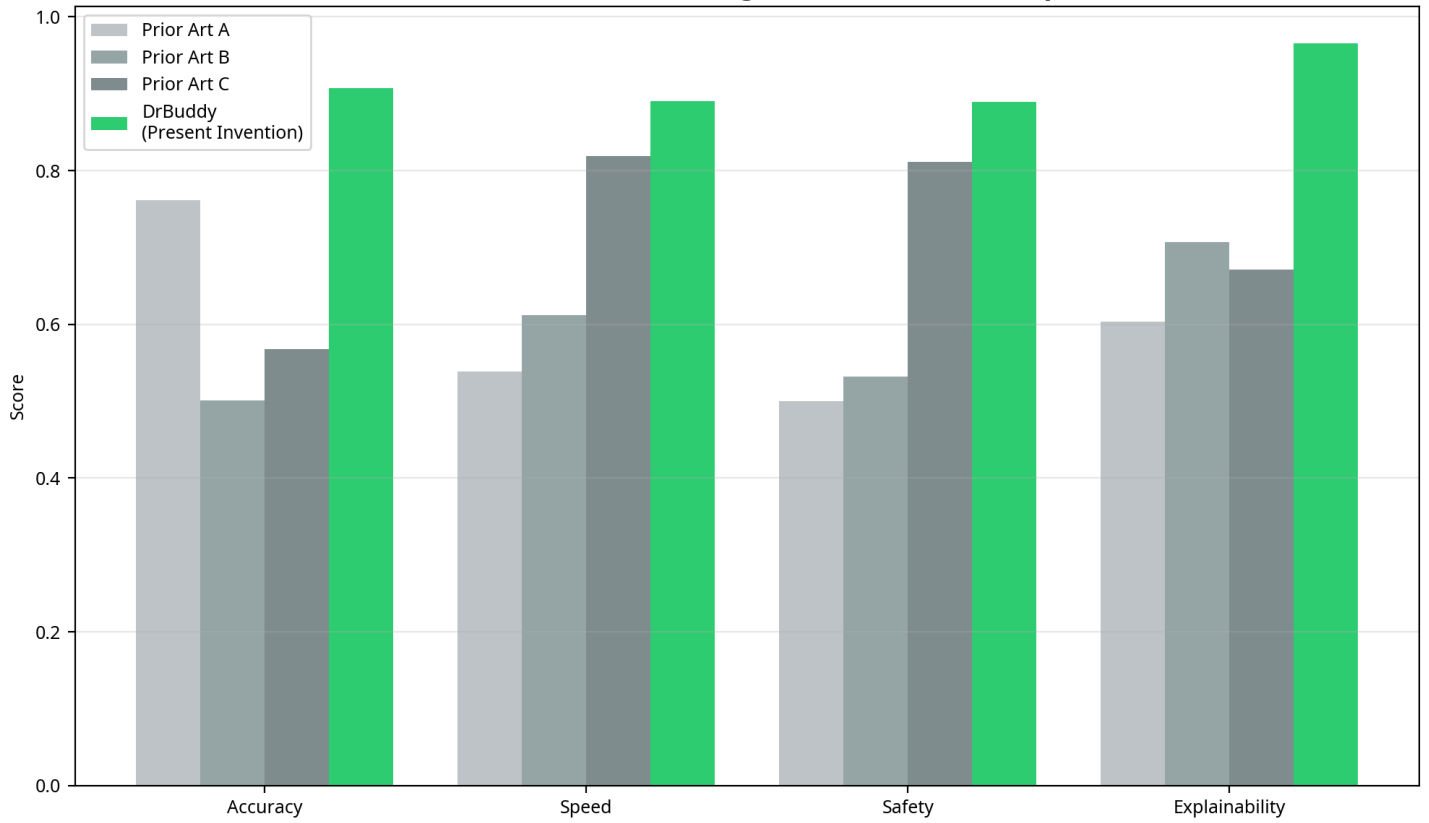


FIG. 6 - PubMed Research Engine Safety Verification Pipeline

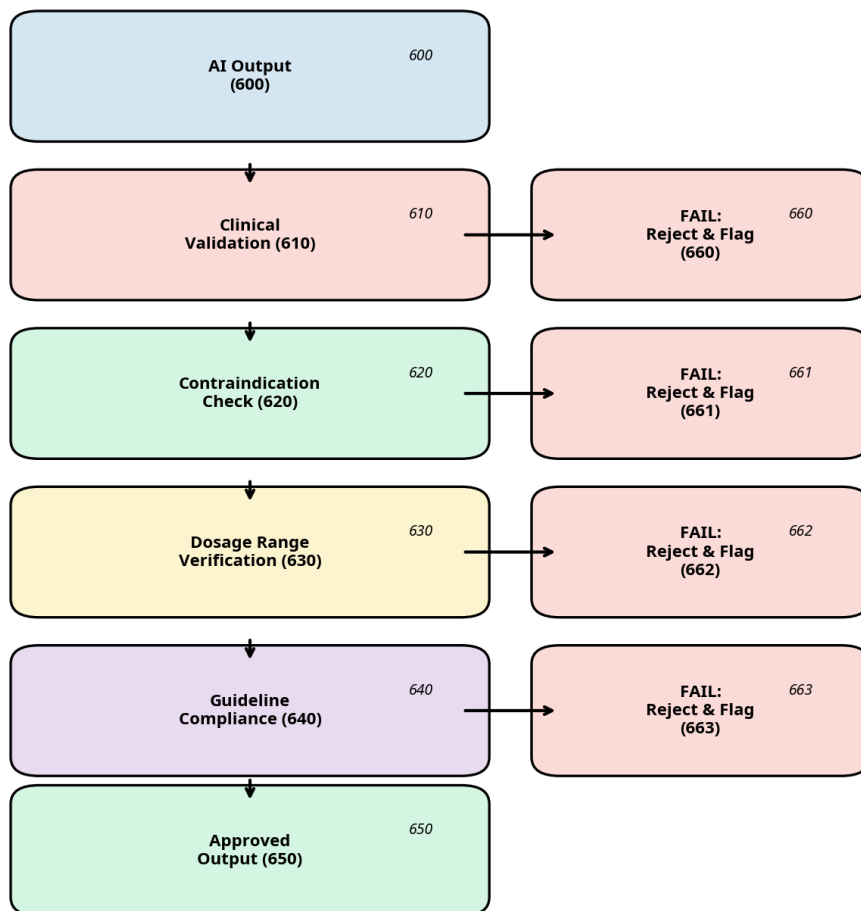


FIG. 7 - PubMed Research Engine Hardware & Cloud Infrastructure

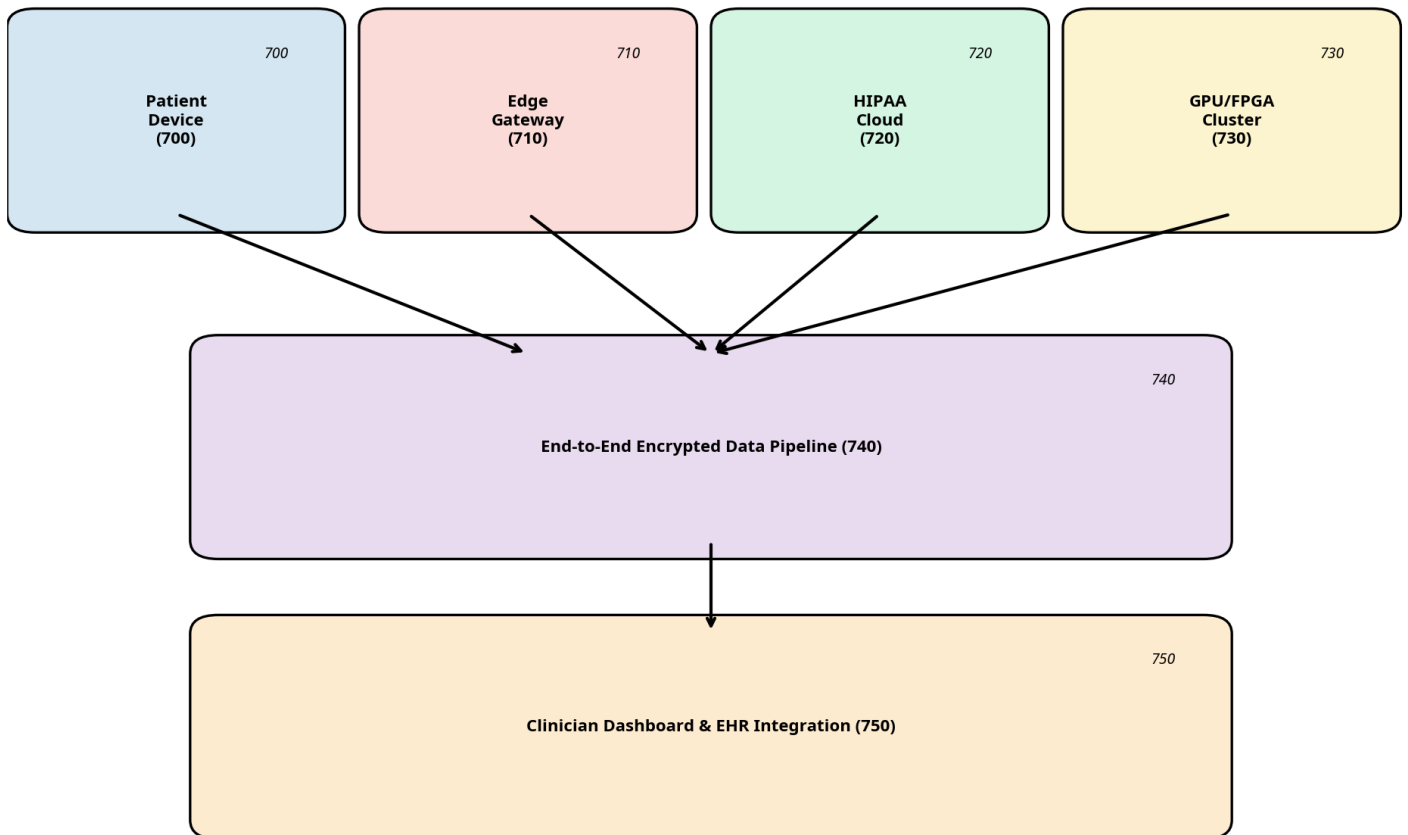


FIG. 8 - PubMed Research Engine Scalability Benchmark

